Price Variation in the Canadian Rental Market

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Abstract. This report looks into the differences in the Canadian rental market, it is based on a dataset with more than 25,000 listings. By focusing on how rent relates to variables like size, bathrooms, bedrooms, furnishing status and pet policies, we are trying to find what correlates most with rent.
Looking into the descriptive statistics and the correlation analysis, we find that the overall property size has the strongest relation to the rent, using the square footage as the strongest correlation, followed by number of bathrooms. We also find that the there is a lot of price variation between the different provinces, when using the median price. This can be a cause of limited listings for some of the provinces within the dataset. We also see a difference in properties that are furnished or not, some property types like Houses and Townhouses have a large difference, both when looking at the absolute and relative price. When it comes to smaller units, the differences are less, some even have no difference at all, like Room For Rent. When it comes to pet policies, the results varies. At the national level, the differences are small, but inside the provinces the it changes a lot, this may indicate that local market conditions matter a lot.

The results show that the rental prices are based on a combination of physical, geographic and local factors, instead of a single dominant variable. Since the analysis is descriptive, the findings should not be used as a cause, but an overview of observable patterns in the dataset.

1 Introduction

Rental prices vary considerably across Canada, but the drivers behind these differences are not always straightforward as demonstrated in this analysis. Factors such as property size, location, furnishing status, and pet policies are all likely to matter. The question is how strongly each of them is associated with rent levels in practice.

This study analyzes more than 25,000 Canadian rental listings to examine how prices differ across diffrent property types, physical characteristics, provinces, furnishing status, and pet policies. The goal is not to explain why prices vary, but to describe the patterns that appear in the data using descriptive statistics and correlation analysis.

The rental prices varies a lot inside Canada, but the reason for this is not always clear as shown in this analysis. Factors like the units size, location, furnishing status and pet policies, all affect the price in combination. The question this analysis ask is how all of these factors are tied together to describe the rental price.

2 Data Preprocessing

2.1 Initial data inspection

The analysis started with an inspection of the dataset, to fully understand the structure, variable types and size. We found 25,350 observations and 18 different variables, having both numbers and text.

Descriptive functions such as `head()` and `info()` were used to explore the data structure, by verifying the data types, and finding missing values. This inspection showed that several variables expected to be numeric, such as square footage, number of bedrooms, and number of bathrooms, but they were stored as text or had inconstant formatting and therefore required cleaning before further analysis.

2.2 Cleaning and Standardizing Square Footage

The variable `sq_feet` was stored as a text variable and contained inconsistent formats. While many observations consisted of a single numeric value, others included ranges (like "700-900"), values with a plus sign ("1000+"), multiple numbers within the same string, or expressions involving multiplication or division (like "11 x 10").

Since these formats cannot be directly interpreted as numerical values, a rule-based cleaning procedure was implemented. By creating a indicator variable to detect ambiguous formats, including values containing ranges, multiple numbers, or mathematical expressions.

Observations identified as ambiguous were excluded from further numerical analysis. For the remaining entries, the numeric components was extracted and converted into a new numerical variable, `sq_feet_num`.

2.3 Logical Filtering of Unrealistic Values

After conversion, the distribution of square footage was examined to identify unrealistic values. Very small or extremely large values were reviewed in relation to property type.

For example, apartment listings below 100 square feet and above 7,999 square feet were considered as errors and were excluded from the analysis. These thresholds were determined through inspection of the distribution and knowledge about realistic property sizes.

A binary variable, `sq_keep`, were created to indicate whether an observation satisfied the different criterias to be valid and if they were going to be included in the further analysis.

2.4 Price per Square feet

To measure the prices between different rental objects with different sizes, price per square feet was the best option, it was calculated based on price devided on square feet:

$$\texttt{price\_per\_sqft} = \frac{\texttt{price}}{\texttt{sq\_feet\_num}}$$

This makes it easier to compare smaller and larger units, since we are comparing how much a fraction of the unit cost and not a direct side by side comparison.

After the variable was made, the distribution was checked to find potential extreme values. Units that had a price above 20 CAD per square feet was removed, since they seemed unrealistic compared to the rest of the dataset. Multiple units had low prices, some had 0, others were below 200 CAD, these numbers were seen as unrealistic, so these were taken out because of incomplete or incorrectly entered information.

2.5 Bedrooms and Bathrooms

`beds` and `baths` were stored as text values, which meant they had to be converted before any calculation could be done.

Some of the listings had "Studio" as bedroom, these were converted into 0 bedrooms, since studio apartments only have one room. Then for the remainig enteries, the numbers were extracted and text was removed, these were saved in a new variable, called `beds_num`.

We did the same with bathrooms, extracting the numbers only and placing it in a new variable. Leaving us with `beds_num` and `baths_num` for the final analysis.

2.6 Furnishing Status

The variable for furnishing had three different categories: "Furnished", "Unfurnished", and "Negotiable". We only used furnished and unfurnished for the analysis, since negotiable does not give any clear directions regarding price, this was excluded.

The rest were stored in a variable called `is_furnished`, this variable is binary, meaning that it is "True" for furnished units and "False" for unfurnished units, making it easier to compare these two in the analysis.

2.7 Pet Policy

When exploring the dataset, we found two columns related to pets, one for dogs and one for cats. These were combined into a single variable called `pet_friendly`, to get a better overview of the units that allows pets.

Listings that had no information regarding either dogs or cats, were treated as not allowing any pets, since permissions are usually stated when they allow it.

2.8 Final Analytical Sample

After adding criterias and cleaning the dataset, the final sample consisted of 20,506 listings. Only observations with valid values for square footage, bedroms, bathrooms and price was used for the final analysis.

To try to prevent values with small observations Property types with listings having under 100 observations where excluded to prevent drawing conclusions from a small amount of groups, for example the median prices from these groups might not be a realistic representation.

3 Data Analysis and Discussion

3.1 Property Type and Price per Square foot

Overall Price Efficiency Across Property Types To compare pricing across property types, median rent per square foot was calculated for each category. Using rent per square foot makes it possible see the differences in size, allowing more direct comparisons between diffrent property types.

The median was used instead of the mean because rental prices in the dataset are somewhat skewed and include extreme values. Using the median therefore provides a clearer picture instead of the mean.

Type	**N**	**Unfurnished**	**Furnished**	**Premium**
Room For Rent	309	0.65	0.62	-0.03
House	1198	1.62	1.84	0.23
Duplex	382	1.68	2.00	0.32
Main Floor	638	1.75	2.24	0.49
Townhouse	1336	1.87	2.12	0.25
Basement	1170	1.88	2.06	0.18
Condo Unit	1795	2.64	3.17	0.54
Apartment	13578	2.84	3.87	1.03

Table 1. Median rent per square foot (CAD) by property type. Premium is calculated as furnished minus unfurnished. N indicates total number of listings per type.

Overall, *Room For Rent* has the lowest median price per square foot. However, since this category represents shared accommodation/apartment rather than independent rental units, it is not directly comparable to the other property types.

Between the rest of the other property types, Houses show the lowest median price per square foot, followed by units like Duplexes and Main Floor. While we find that, Apartments and Condo Units, on the other hand, have the highest median price per square foot. This indicates that more compact units, typically found in the middle of cities, tends to have higher prices relative to the space offered.

These results suggest that units found on the ground, like houses, may provide more space for the price compared to vertical housing formats like apartments. At the same time, these differences may also indicate variation in location, building age, amenities, and target market across the different property categories.

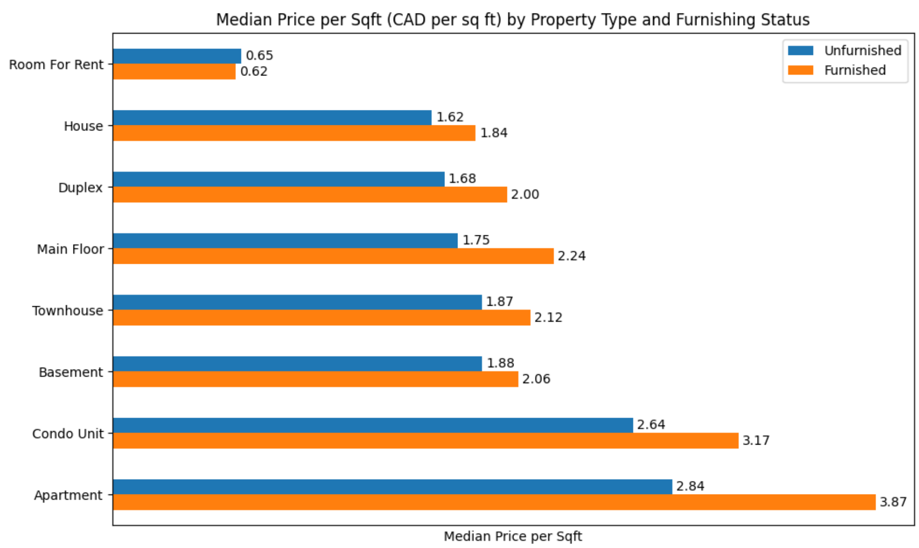

Fig. 1. As illustrated in this Figure, furnished listings shows a higher price per square foot across most of the property types.

When looking at the figure, we can see that the furnishing premium is not the same across all the diffrent property types. With apartments showing the largest difference in price per square foot between the furnished and the unfurnished units. We see a similar pattern for Condo Units and Main floor, that there is a clear increase in price when the units are furnished.

While in Houses and basements, the difference is smaller, *Room For Rent* have almost no price gap between the furnished and unfurnished, this could be because shared accommodations often includes basic furniture like beds and other types of furniture.

Interpretation of the Furnishing Premium The furnishing premium is most noticeable in smaller and typically urban housing formats such as Apartments and Condo Units. These units may attract tenants who are more flexible or looking for "move-in-ready" options, which could increase the willingness to pay extra for furnished listings.

In contrast, larger property types such as Houses may be rented by longer term tenants who prefer to furnish and decorate the space themselves. This could explain why the furnishing premium is smaller for these properties because of lower demand of furnished units.

Overall, the results show that both property type and furnishing status are linked to differences in price per square foot. However, these relationships should not be interpreted as causal, since furnishing decisions may also be related to location, property quality, or the type of tenants targeted.

In summary, shared accommodation(Room For Rent) has the lowest median price per square foot overall, while Houses appear to offer the most space efficient option among full residential units. Furnishing adds a measurable premium in most property types, particularly Apartments, which may reflect different demand patterns across the different property types.

3.2 Physical Dimensions and Rental Price

Correlation Between Physical Dimensions and Price To better understand the relationship between rental price and the physical characteristics, Pearson correlation coefficients were computed for price, square footage, number of bedrooms, and number of bathrooms.

Pearsons correlation coefficient measures the strength and direction of a linear relationship between two continuous variables. The coefficient ranges from -1 to 1, where values closer to 1 (+ or -) indicates a stronger linear association.

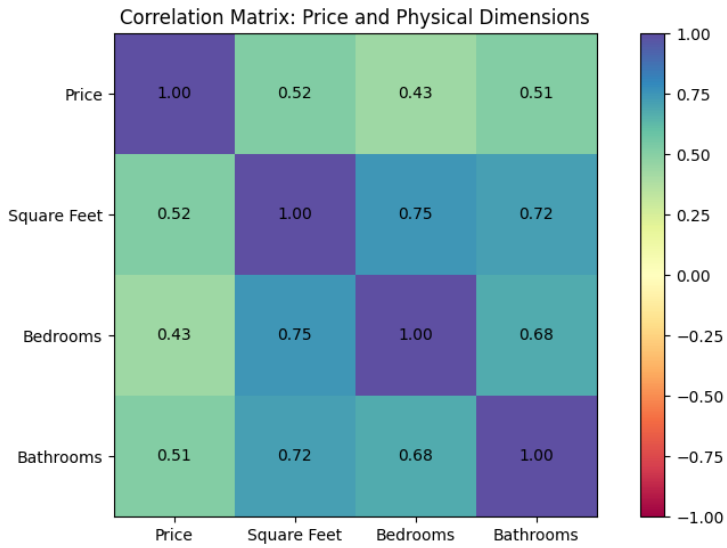

Fig. 2. As shown in this Figure, all three physical dimensions demonstrate positive correlations with rental price.

Square footage shows the strongest correlation with price (r = 0.52), followed closely by the number of bathrooms (r = 0.51). The number of bedrooms has a weaker, but still moderate and positive correlation (r = 0.43).

Interpretation of Results The results indicates that larger properties generally have higher rental prices. Between the variables examined, square footage shows the strongest relationship with the price.

We can see that bathrooms has almost the same correlation with price as the square footage. This could be because that properties with more bathrooms are often larger and may have a higher quality compared to similar sized units or the age, since newer units tends to have more bathrooms.

And bedrooms have the weakest correlation with price compared square footage and bathrooms. This indicates that adding more bedrooms alone does not really increase rent unless it also comes with more space or more bathrooms.

Interrelationship Between Physical Variables When looking at square footage, bedrooms, and bathrooms, we see that they are also closely related to each other. For example, square footage has a correlation of 0.75 with bedrooms and 0.72 with bathrooms, this indicates that size often comes with more rooms and bathrooms.

Because these variables usally increases together, they describe more or less the same result of the property size, this makes it hard to isolate the individual impact each variable have on the rent price without analyzing them together.

Since the relationships are positive and shows correlation, none of the variables clearly stands out alone, this makes it hard to give a clear answer to the rental price. This might suggest that other factors like location, furnishing status or property type also plays an important role regarding the price.

3.3 Pet-Friendly Premium Analysis

National-Level Pet Premium The median rental price was compared between pet friendly and non pet friendly listings at the national level to examine if allowing pets is associated with higher rents.

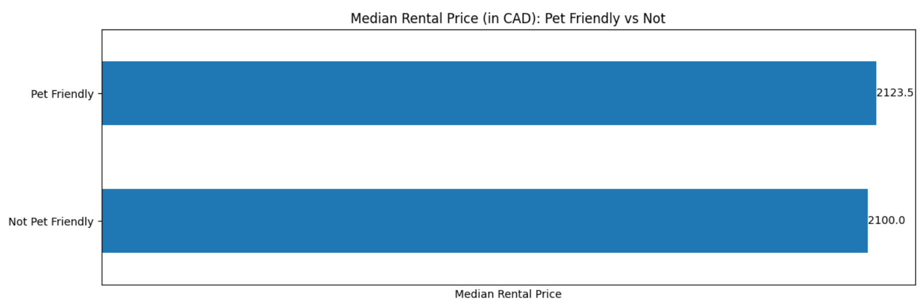

Fig. 3. As shown in this Figure, the median rent for pet friendly listings (2123.5 CAD) is slightly higher than for non pet friendly listings (2100 CAD), resulting in a national-level pet premium of around 23.5 CAD per month.

Since the difference is small, allowing pets does not appear to have a strong impact on rental price at the national level.

Provincial Variation in Pet Premium However, when the data is looked at the provincial level, the pattern changes.

Province	N (Pet)	N (No Pet)	Median (Pet)	Median (No Pet)	Premium (CAD)
New Brunswick	2	7	2032	1640	392
Manitoba	661	106	1700	1409	290
Quebec	1552	304	2050	1802	248
Ontario	4600	966	2489	2275	214
Newfoundland and Labrador	6	3	1180	1020	160
Nova Scotia	181	28	2425	2400	24
Alberta	6098	4295	1995	2000	-5
British Columbia	808	239	2300	2600	-300
Saskatchewan	516	34	1305	1799	-494

Table 2. Median rental price comparison between pet friendly and non pet friendly listings by province, including sample sizes(N).

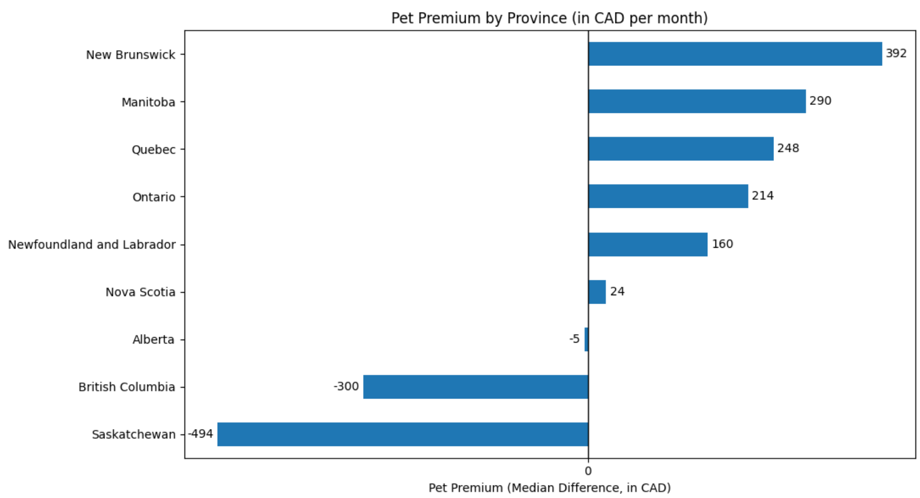

Fig. 4. This Figure illustrates that the pet premium have big variations between provinces.

By looking at individual provinces, the pattern varies noticeably. For example, *New Brunswick* (+392 CAD) and *Manitoba* (+290 CAD) show a relative high pet premium. In contrast, *Saskatchewan* (-494 CAD) and *British Columbia* (-300 CAD) show a negative difference, meaning that pet-friendly listings in these provinces have lower median rents.

Looking at the provinces separately, there is no clear pattern linked to the findings we saw at the national level. In some regions, pet-friendly listings have higher median rents, while in others the difference is negative. This shows that the relationship between pet policies and rent changes depending on the local market conditions.

When looking at the sample size for each provinces, we see that some variables like New Brunswick and Newfoundland and Labrador have few observations. This can affect the median price by changing a lot when new listings are added, this can result in unfair comparisons, since the results might be bias because of limited data.

Another factor that might tell a different story is the property type, since Pet-friendly units might be more normal for larger properties like houses, with a higher rental price compared to smaller apartments. Age of the units might also matter, newer buildings might not allow pets because of higher wear and tear and newer apartments tend to be more expensive. Because of this, the observations for pet premium should only be related to the patterns in the dataset and not a cause for increase or decrease in rent price.

3.4 Distribution of Rental Prices Across Provinces

Median Rental Prices by Province A boxplot graph is used to compare rental prices between the different provinces, together with a statistical summary for providing additional context.

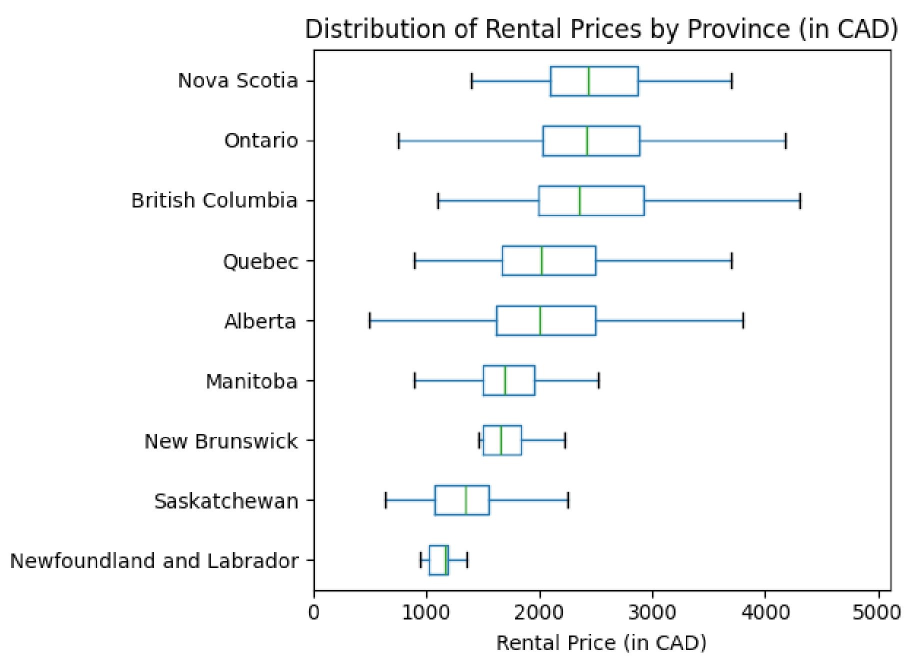

Fig. 5. This Boxplot Figure highlights the distribution of rental prices by province. Median rental prices vary a lot across the different regions.

Nova Scotia shows the highest median rent in the dataset by 2425 CAD per month together with Ontario (2417 CAD) and British Columbia (2360 CAD). While Newfoundland and Labrador have the lowest median rent of only 1165 CAD, this highlights how big difference the price ranges is from province to province. This can indicate that some of the provinces are more populated and have bigger cities which tends to have a higher rent.

Variability and Dispersion To look closer at the price variation for each province, a method calculating the interquartile range (IQR) was used. The IQR measures the spread of the middle 50% of the observations and is calculated as:

$$\text{IQR} = Q_3 - Q_1$$

The Q_1 represents the lower 25th percentile and Q_3 the upper 75th percentile of the rental prices. Because, the IQR(50%) only the central portion of the distribution, it is less affected by extreme values compared to measuring range or the mean. [1].

Province	**N**	**Q1**	**Median**	**Q3**	**IQR**
Newfoundland and Labrador	9	1020	1165	1195	175
Saskatchewan	550	1075	1347	1554	479
New Brunswick	9	1505	1655	1835	330
Manitoba	767	1505	1695	1950	445
Alberta	10393	1619	2000	2490	871
Quebec	1856	1675	2020	2495	820
British Columbia	1047	1995	2360	2925	930
Ontario	5566	2030	2417	2889	859
Nova Scotia	209	2100	2425	2875	775

Table 3. Summary of rental the price distribution by province, including sample size (N), first quartile (Q1), median, third quartile (Q3), and interquartile range (IQR), measured in CAD per month.

The variation in price is not the same across the provinces as we can see in the table. British Columbia has the largest IQR at 930 CAD per month followed by Alberta (871 cad) and Ontario (859 CAD). This shows a bigger spread in these local markets, suggesting that it might be because of larger listings available or that these provinces may have larger cities with higher rent prices, which can increase the price variation.

If take a look at Newfoundland and Labrador (175 CAD) and New Brunswick(330 CAD), we see that both of them have only nine listings, compared to the provinces with the highest variations, this might indicate that the results are less stable and can change when the size is small.

Distribution Shape and Outliers To make the boxplots easier to read, values above the 99th percentile were excluded from this figure. Since extreme values were removed from the boxplots for readability, they were still kept in the summary table, and including in the IQR calculations.

In several provinces, the price distribution is skewed to the right. This means that a small number of high priced listings extend to the upper tail. As a result of this, the median provides a more representative summary measure than the mean when comparing provinces.

3.5 Furnishing Premium Across Property Types

The furnishing premium was calculated by comparing the median rent of furnished and unfurnished units within each property type. It is calculated as the difference between the two median values like this:

$$\text{Premium} = \text{Median}_{\text{Furnished}} - \text{Median}_{\text{Unfurnished}}$$

Type	**N (Furnished)**	**N (Unfurnished)**	**Premium (CAD)**
Townhouse	81	1255	845
House	137	1061	600
Condo Unit	422	1373	470
Duplex	39	343	275
Main Floor	50	588	200
Apartment	408	13170	185
Basement	169	1001	50
Room For Rent	254	55	0

Table 4. The table shows the number of furnished and unfurnished listings (N) and the difference in median monthly rent (CAD) between the two groups, finding the Furnishing Premium.

We can see a clear difference between the different property types. Townhouses has the largest furnishing premium with 845 CAD per month, together with Houses(600 CAD) and Condo units(470 CAD). Basement on the other scale, only have a premium of 50 CAD, while Room For Rent not having a difference at all between furnished and unfurnished units.

When looking at the sample sizes, we can see that furnished have fewer listings than unfurnished, but most property types have enough observations to support the findings. But Duplex and Main floor units, have less than 60 furnished listings, so these results should be interpreted with caution.

Relative Furnishing Premium (%) Because property types with higher rents will show larger absolute gaps, the furnishing premium was also calculated in percentage, to see if the result changes.

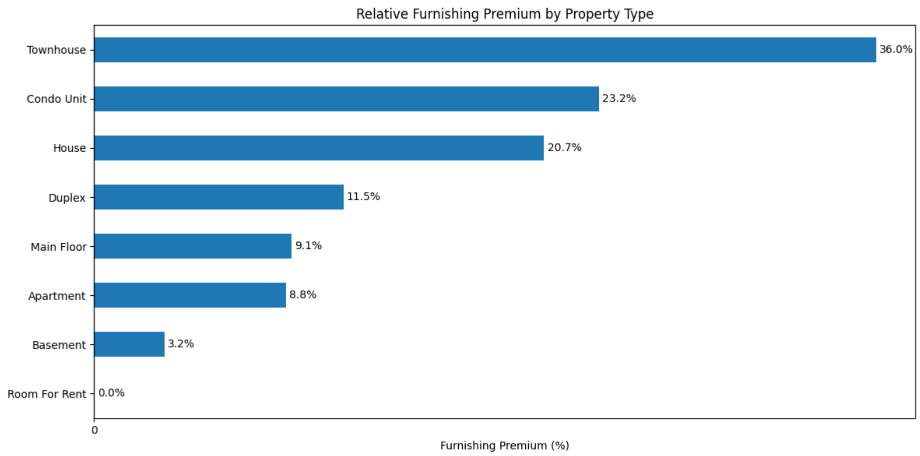

Fig. 6. Furnishing Premium by Property Type in Percentage

Looking at the figure, Townhouses displays the biggest relative furnishing premium at around 36% and Condo Units (around 23%) both displaying a clear difference in furnished vs unfurnished. For Basements we only see a increase in around 3%, while Room For Rent, shows no difference at all.

This strongly suggest that the differences are clearly based on property types and potential tenant groups they attract, furnished houses might attract renters seeking short term rent(for example, work relocation or during renovations), while Apartments might attract people who seek long time rent (For example, younger people).

Finally, the results shows that furnising premium varies bot in absolute and relative terms, with the strongest effect in Townhouses and houses, but smaller and shared units show minimal differences between furnished and unfurnished listings. This indicates that property type and different demands in the markets, decides the differences in price.

3.6 Discussion

The results shows that rental prices differ between diffrent property types, physical characteristics, and provinces in this dataset. Prices are not evenly distributed, and multiple diffrent factors appear to contribute to the patterns found in the dataset.

Starting with the structural characteristics, where square footage shows the strongest correlation with the rent (r = 0.52), followed by the number of bathrooms (r = 0.51). While bedrooms have a weaker relationship compared to the other two. This indicates that overall size and layout might matter more than just the number of rooms per unit. It could also indicate that newer units tends to include both more bathrooms and larger living areas, which may explain the stronger correlation with the rental price.

We can also see clear regional differences when comparing the diffrent provinces. Nova Scotia, Ontario, and British Columbia have a higher median rents in the dataset, while provinces such as Newfoundland and Labrador and Saskatchewan have a lower. These provinces might have larger cities, with higher rents compared to the others. The price variations inside the provinces also differs compared with eachother. British Columbia, Alberta, and Ontario shows relative wide interquartile ranges, meaning that rents vary more inside these markets. Compared to the smaller provinces with less spread, this might be influenced by smaller sample sizes and also differences in property types/local variables.

The furnishing premium is not consistent across the different property types. Looking at Townhouses and Houses that shows the largest differences between furnished and unfurnished units, both in absolute and relative terms. While smaller units, like Basements and Room for Rent listings, show a limited price differences in both absolute and relative terms. This indicates that the value of furnishing depends on the property type and potentially the type of renters these units may attract creating different demands.

While the findings for pet policies are less clear. Looking at the the national level, the difference between pet friendly and non pet friendly listings are small by only 23.5 CAD. However, when looking at individual provinces, the pattern changes alot. For example, in some regions like New Brunswick, pet friendly listings are related with higher rents, while in others like Saskatchewan they have lower median rents. This variation may indicate that the pricing effect of allowing pets likely depends on local market conditions and the diffrent kinds of properties that may tend to allow for pets.

Finally, we see that several provinces have right skewed price distributions, where a small number of high priced listings stretch in the upper end of the distributions. Because of this, using the median provides a more stable and better measure than the mean when comparing.

To summarize the results of this analysis, suggest that rental prices reflect a combination of physical features, regional differences, and different local markets. No single variable fully explains price differences alone in the dataset.

Limitations and Potential Sources of Bias However, there are some limitations that needs to be kept in mind when interpreting the results of this analysis.

The prices in this dataset are based on asking prices, rather than actual transaction prices, this may paint a different picture than the real world. Meaning that the final rent can differ from the price reported in the dataset, due to negotiation, marked demand or other changes in the market. This means that the analysis reflects advertised prices, and not confirmed rental agreements based data representing the actual market.

Representation is another issue, when the number of listings varies a lot between provinces and property types. Looking at provinces with few observations, the results may appear more stable or uniform because there is less data available, compared to the provinces with more observation. This makes the comparisons between provinces uneven and can give bias results.

While the analysis is descriptive and primarily only checks relationships between two variables at a time. Measures like medians and interquartile ranges provide useful information about distribution, but they do not include relevant factors like quality of the neighborhood, population density, or average income. Because of this, the results should be seen as patterns in the data rather than any explanations.

When having the Furnishing status as a binary yes and no variable in the dataset, some valuable information might be lost. In reality, furnishing can vary widely in quality and appearance, making it harder to understand a concrete price difference when comparing two similar units. Furnished units can differ widely, so the calculated premium reflects a broad average rather than a detailed picture.

Lastly, some listings with high values above the 99th percentile were excluded from the visualizations to improve readability. Even though, these observations were still included and used in the numerical calculations and summary table, removing them from the figures might affect how the price dispersion are interpreted visually.

Overall, the analysis describes patterns within this Canadian Rental Market dataset. The findings should not be interpreted as complete explanations of how rental prices are set in the rental markets outside the dataset.

4 Conclusion

In conclusion, this report examined how rental prices vary across diffrent property characteristics, provinces, furnishing status, and pet policies by using descriptive statistics and correlation analysis.

The results highlights that structural characteristics like size matter. Square footage showed the strongest correlation with rent, followed by the number of bathrooms, while the bedroom count showed a weaker relationship than the other two. This indicates that the overall size and layouts are more important than the number of rooms. It might also suggest that newer units often combine larger floor areas with more bathrooms compared to older ones.

Regional differences was also found when comparing the provinces. We found that median rent levels and price variation differ between provinces, indicating that rental markets operates differently based on location. However, these differences should be viewed with caution, as the number of listings differs between provinces, making some of the values less stable.

The results related to the pet policies are less consistent when looking at national vs provinces. While the national level difference are smaller, provincial comparisons reveal noticeable variation in the results. For example, in some regions, pet-friendly listings are related to higher rents, while in others we find them being cheaper. This indicates that the relationship between pet policies and rent varies by location or another unknown reason, rather than following one consistent pattern found in the dataset.

Moving on to the furnishing premium as this is not uniform across the diffrent property types. Houses and Townhouses show stronger differences between furnished and unfurnished units, both in absolute terms and as percentages compared to the others. Smaller or shared units only show a small price differences between furnished and unfurnished listings. This might suggest that the value of including furnishing depends on property type and the different markets these properties are in.

Overall, this analysis finds that rental prices are influenced by a combination of structural and regional factors, as well as differences in who the properties are intended for. Not single factor can explain the observed variation alone.

Since this analysis is descriptive, the findings should be seen as patterns in the data rather than proof of cause and effect. But, the analysis gives a clear overview of how prices vary and can be used as a starting point for future analysis.

References

1. The MathWorks, Inc.: iqr. MATLAB Documentation, https://www.mathworks.com/help/matlab/ref/iqr.html